DAY 08 OF 16 · 3D & 6DOF STREAMING

Cleaning & aligning scans

Real scans are noisy and come in pieces. Today we clean them and align them, and discover that alignment means finding a 6DoF pose.

Noise & outliers · Downsampling · Normals · Registration · ICP

Book: Ch. 5–6 · Code: day08_cleaning_registration/

BOOK CH. 5–6 · PREPROCESSING

Cleaning: less but better points

Voxel downsampling

Keep one point per small cube. Even density, fewer points (37k → 29k at 5 cm).

01

Statistical outlier removal

Drop points whose mean neighbour distance is unusually large (2,000 fake outliers → 1,627 removed).

02

Normals

Fit a plane to each neighbourhood; its perpendicular is the normal. Needed for meshing, lighting and ICP.

03

BOOK CH. 6 · REGISTRATION

Registration = find the pose that aligns two scans

$$\text{find } R,\ t \text{ minimising } \Sigma \ \| R \cdot p_i + t - q_i \|^2$$

p_i

Points of the scan we move (source).

q_i

Matching points in the reference (target).

$R,\ t$

The 6DoF pose: exactly the matrix from Day 2.

BOOK CH. 6 · ICP

Iterative Closest Point in four repeated steps

01

Match

For each source point, take the closest target point (k-d tree!).

02

Solve

Compute the R, t that best aligns the matches.

03

Apply

Move the source by that pose.

04

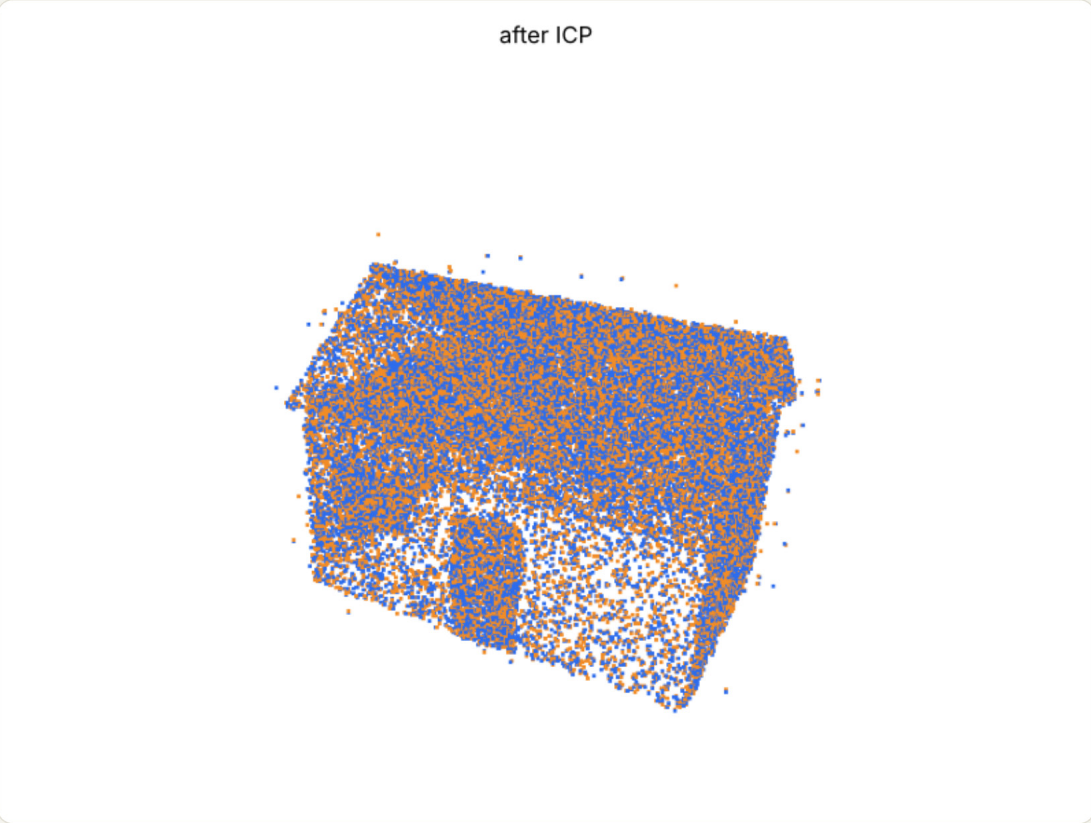
Repeat

Until the error stops falling.

Each step uses something from earlier sessions: k-d trees (Day 7), poses (Day 2), normals (today).

LAB RESULT

ICP recovers a secret 6DoF pose



- Before: overlap 0.56, RMSE 0.167 m
- After: overlap 1.00, RMSE 0.017 m
- Point-to-point got stuck; point-to-plane did not
- Same idea tracks headsets and robots (SLAM)

Secret pose: 12° yaw, 3° pitch, −2° roll, (0.8, −0.5, 0.2) m. Point-to-plane ICP error: 0.0 cm and 0.00°.

HANDS-ON LAB · COPY, RUN, LOOK

Clean a noisy scan and find the hidden pose

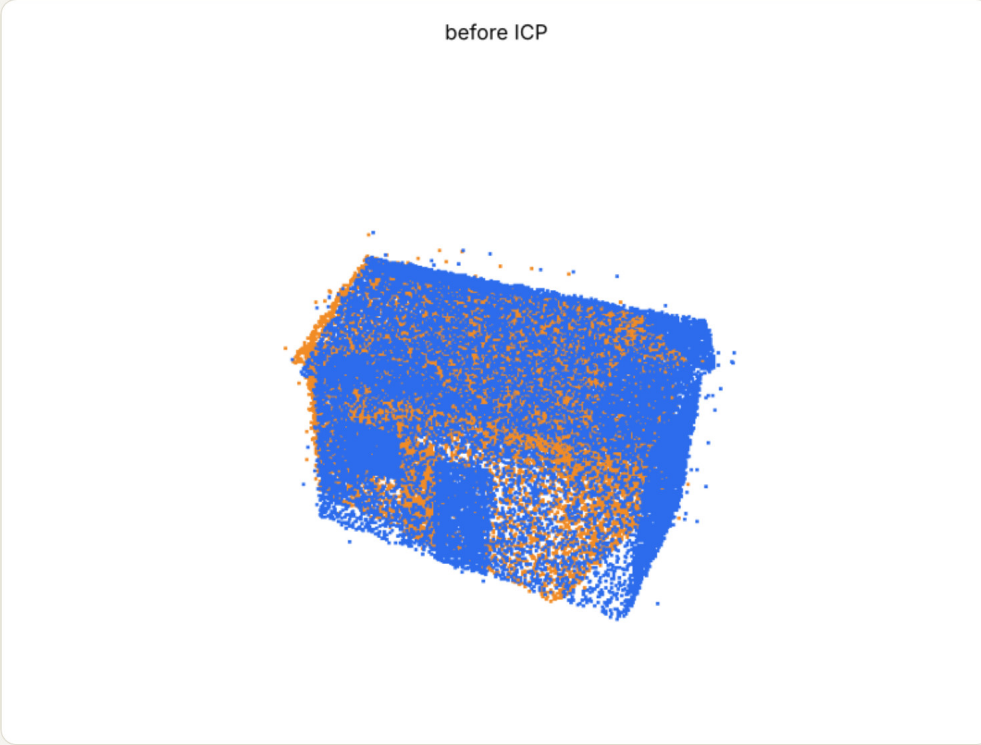
RUN IT

```
cd day08_cleaning_registration
python clean_and_register.py
```

→ Four windows: raw, cleaned, before ICP, after ICP.

→ Compare the secret and recovered matrices.

→ Switch to PointToPoint and rerun.



Three things to remember

1. Downsample and remove outliers before anything else.
2. Registration finds the 6DoF pose that aligns two scans.
3. ICP = match closest points, solve the pose, repeat. It needs a good first guess.

BEFORE NEXT SESSION

Finish Lab L4. Next: from geometry to meaning.